

XIAMEN UNIVERSITY

Programming Language C

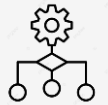
Week 3 - Chapter 3 (i)

3.1 Branching and Jumps

Lab

Dr. Najla'a

Chapter 3 Control Flow



- **3.1 Branching and Jumps**
  - 3.1.1 if statements
  - 3.1.2 switch...case Statement
- **3.2 Loop**
  - 3.2.1 while Loop
  - 3.2.2 do .. while Loop
  - 3.2.3 for Loop
- 3.3 Nested Loops

Exercise 1

- Ex.1 Write a program that reads an integer and determines and prints whether it's odd or even. [Hint: Use the remainder operator. An even number is a multiple of two. Any multiple of two leaves a remainder of zero when divided by 2.]

Note: use "if" statement.

Exercise 1: Mini Parking Fee

Goal: Compute fee with one override and one simple discount.

Inputs (single line):
hours, type, lostTicket

- hours = 1..24 (int)
- type = C (car) or M (moto)
- lostTicket = 0/1

Rules (apply in this order):

1. If lostTicket==1 → fee = 50.
2. Else base: Car RM 4/h, Moto RM 2/h.
3. If hours >= 6 → 20% discount on the fee.
4. Minimum payable RM2 (unless computed fee is 0).

Sample: 7 C 0 → base 28 → 20% off = 22.4 → print RM22.40

Exercise 1: Solution

```c
#include <stdio.h>

int main(void) {
    int hours, lostTicket;
    char type;
    double fee = 0.0, rate = 0.0;

    printf("hours type(C/M) lost(0/1): ");
    if (scanf("%d %c %d", &hours, &type, &lostTicket) != 3) return 0;

    if (lostTicket == 1) {
        fee = 50.0;
        printf("Lost ticket.\n");
    } else {
        if (type == 'C') rate = 4.0;
        else if (type == 'M') rate = 2.0;
        else { printf("Invalid type\n"); return 0; }

        fee = hours * rate;

        if (hours >= 6) fee *= 0.8;          // 20% discount
        if (fee > 0.0 && fee < 2.0) fee = 2.0;  // minimum payable
    }

    printf("RM%.2f\n", fee);
    return 0;
}
```

Exercise 2: Elevator Quick Check

Goal: Decide the elevator action.

Inputs (single line):

loadKg, limitKg, doorClosed, fireAlarm

- doorClosed = 0/1, fireAlarm = 0/1

Rules (priority order):

1. If fireAlarm==1 → print "LOCKED_FIRE".
2. Else if doorClosed==0 → print "BLOCKED_DOOR".
3. Else if loadKg > limitKg → print "BLOCKED_OVERLOAD".
4. Else if loadKg > 0.9*limitKg → print "CAUTION_MOVE".
5. Else → print "**MOVE".**

Sample: 720 800 1 0 → CAUTION_MOVE

Exercise 2: Solution

```c
#include <stdio.h>

int main(void) {
    int loadKg, limitKg, doorClosed, fireAlarm;
     printf("load, limit, door(0/1), fire(0/1): ");
    if (scanf("%d %d %d %d", &loadKg, &limitKg, &doorClosed, &fireAlarm) != 4) return 0;

    if (fireAlarm == 1) {
        printf("LOCKED FIRE\n");
    } else if (doorClosed == 0) {
        printf("BLOCKED DOOR\n");
    } else if (loadKg > limitKg) {
        printf("BLOCKED OVERLOAD\n");
    } else if (loadKg > (int)(0.9 * limitKg)) {
        printf("CAUTION MOVE\n");
    } else {
        printf("MOVE\n");
    }
    return 0;
}
```

Exercise 3: Tiny SMS Risk (LOW / HIGH)

Goal: Classify an SMS with two simple checks.
Inputs (single line):
hasLink, unknownSender, urgentMoney

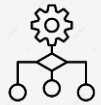
- Each is 0/1
- **urgentMoney** means “mentions urgency AND money” already combined.

Rules (in order):

1. If hasLink==1 **and** unknownSender==1 → "HIGH".
2. Else if urgentMoney==1 → "HIGH".
3. Else → "LOW".

Sample: 1 1 0 → HIGH
Sample: 0 1 1 → HIGH
Sample: 1 0 0 → LOW

Exercise 3: Solution



```c
#include <stdio.h>

int main(void) {
    int hasLink, unknownSender, urgentMoney;
    printf("link(0/1) unknown(0/1) urgentMoney(0/1): ");
    if (scanf("%d %d %d", &hasLink, &unknownSender, &urgentMoney) != 3) return 0;

    if (hasLink == 1 && unknownSender == 1) {
        printf("HIGH\n");
    } else if (urgentMoney == 1) {
        printf("HIGH\n");
    } else {
        printf("LOW\n");
    }
    return 0;
}
```